HOMEWORK 8.5 · UNIT 8 · NC.8.G.4 · NC.8.G.3 · NC.8.G.2

Dilations Day 3

Apr 19 / Apr 20 · Targets: G.4-3, G.4-4, G.4-2, G.3-4, G.4-5, G.2-5, G.4-6, G.4-8

Name _____ Date _____ Block _____

Remember: Similar figures (~): matching angles are equal, and matching sides all have the same ratio (the scale factor). Order matters: △ABC ~ △DEF means A ↔ D, B ↔ E, C ↔ F. **Example: sides 3, 4, 5 and 6, 8, 10:** $\frac{6}{3} = \frac{8}{4} = \frac{10}{5} = 2$, **so they are similar.**

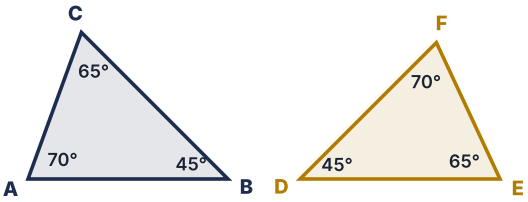
SCORE 3.0 Approaching · the basics

Targets: G.4-3, G.4-4, G.4-2, G.3-4

1 A triangle has sides 4, 6, and 8. Which triangle is similar to it? G.4-3

(A) 6, 9, 12
(B) 5, 7, 9
(C) 8, 12, 20
(D) 2, 4, 6

2 △ABC is similar to △DEF. Which statement is correct? G.4-4



(A) △ABC ~ △FDE
(B) △ABC ~ △DEF
(C) △ABC ~ △EFD
(D) △ABC ~ △FED

3 A triangle with sides 12, 16, 20 is dilated to sides 3, 4, 5. Find the scale factor. G.4-2

_____

_____

4 Is a 3 by 5 rectangle similar to a 9 by 15 rectangle? G.4-3

(Yes) (No)

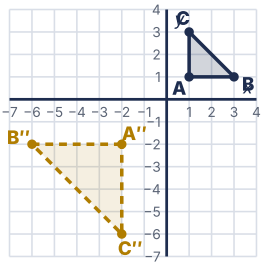
Why? _____

5 Dilate (−2, 6) by k = 1.5 about the origin. REVIEW 8.3 G.3-4

_____

SCORE 4.0 Proficient · on target Targets: G.4-5, G.2-5, G.4-6

6 Which sequence maps △ABC onto △A″B″C″? G.4-5



Ⓐ Dilate by 2, then rotate 180°
Ⓑ Dilate by 2, then reflect over the x-axis
Ⓒ Dilate by $\frac{1}{2}$, then rotate 180°
Ⓓ Rotate 180°, then translate 1 left and 1 down

7 A 90° rotation followed by a translation makes a congruent figure. REVIEW 8.2 G.2-5

True False

8 **Lin** made a mistake. Circle it. Explain it. Fix it. REVIEW 8.3 G.4-6

LIN'S WORK
The rule $(x+2, y+2)$ makes a bigger, similar figure.

SCORE 5.0 Mastery · go deeper Targets: G.4-8

9 △ABC has A(1, 2), B(2, 1), C(3, 2). △DEF has D(2, −4), E(4, −2), F(6, −4). Prove △ABC ~ △DEF with a sequence of transformations. G.4-8

First _____ . Then _____ . The triangles are similar because _____ .

10 Why do similar figures have equal angles but sides that are proportional (not equal)? G.4-8

The angles stay the same because _____ . The sides are proportional because _____ .